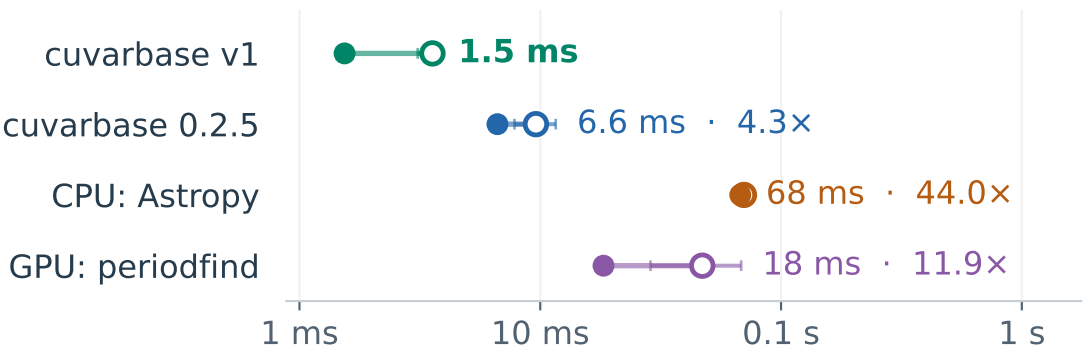


Faster transit searches across TESS and ZTF cadences

Search time per lightcurve · lower is faster

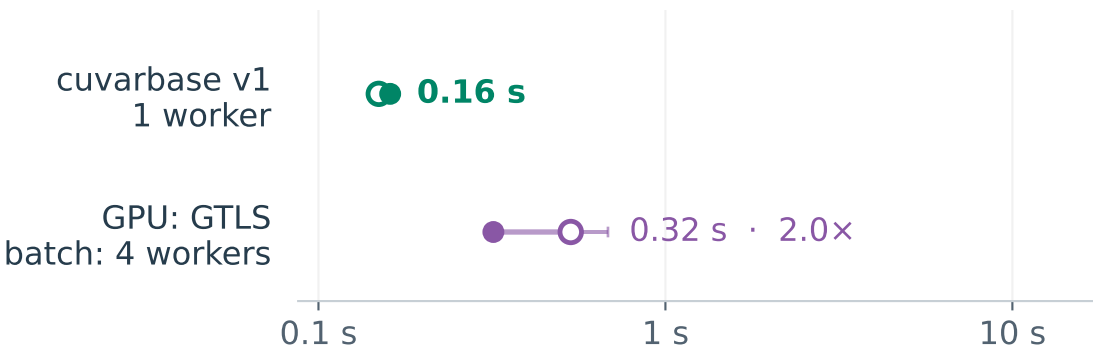
○ One lightcurve ● Batch of 16: time per lightcurve
BLS / TESS: one dense sector

200 s cadence · up to 9,736 samples · 26 days



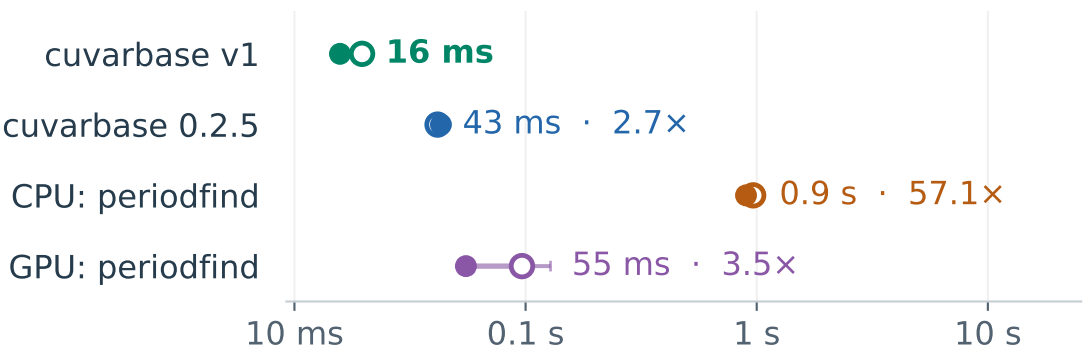
TLS / TESS: one dense sector

9,736 samples · 26 days · 2,325 trial periods



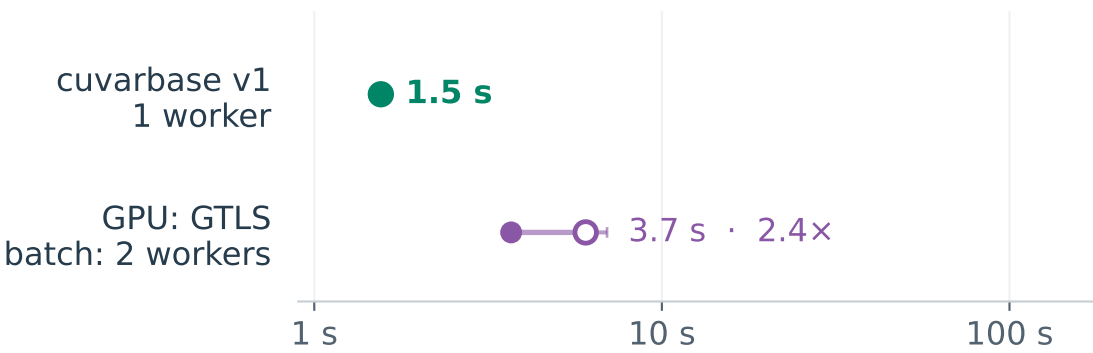
BLS / TESS: two separated sectors

30 / 10 min cadence · up to 4,295 samples · 735 days



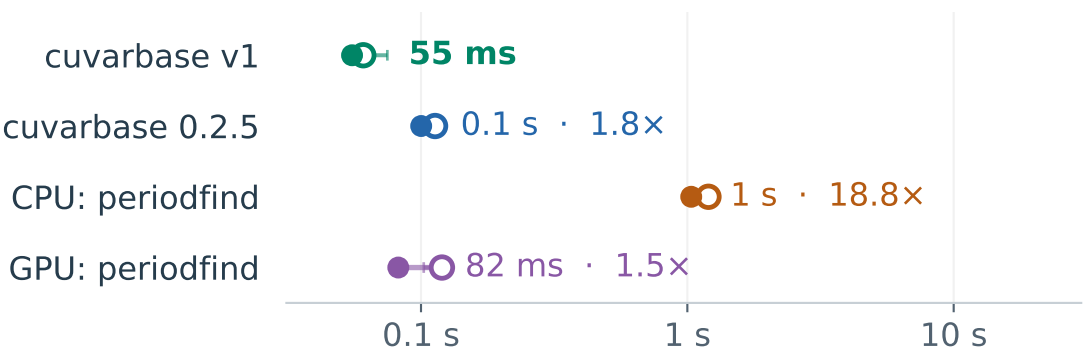
TLS / TESS: two separated sectors

4,295 samples · 735 days · 74,616 trial periods



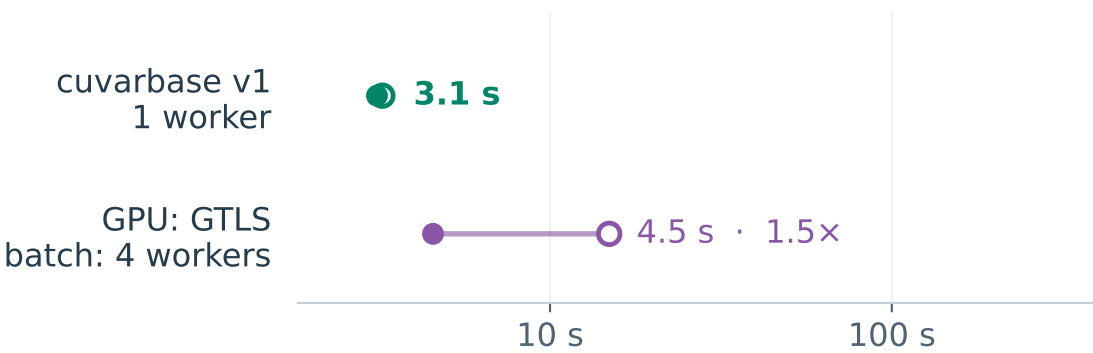
BLS / ZTF: sparse g/r

Up to 1,317 samples · 2,744 days



TLS / ZTF: sparse g/r

1,317 samples · 2,744 days · 235,266 trial periods



Labels give batch time and the ratio to v1. Medians of 5 single / 3 batch calls; Whiskers span repetitions; logarithmic axes.

BLS: A40; TLS: RTX A6000. Warm APIs; input loading and grid construction excluded. CPU allocations: see report.

GTLS batch: fastest eligible 1/2/4-worker pool. Recovery qualifications and search/diagnostic timings: see report.

Post hoc report of complete configurations: original campaign gate failed after 4-worker GTLS ran out of memory on separated TESS.